Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Digital Logics

BCA — Semester 1 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Design a BCD-to Seven-segment decoder to display decimal numbers 1, 2 and 4.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What are the drawbacks of clocked RS flip flop? Explain the operation of a JK flip flop along with its characteristic table, characteristic equation circuit diagram and timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Design a sequential circuit using JK flip-flop with the help of given state diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Convert (110.101)8into binary and decimal number system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Subtract (739.57)■■— (78.35)■ using both 10's and 9's complement.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is magnitude comparator? Design 2-bit magnitude comparator.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define Half-subtractor with truth table and logic diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is decoder? Implement 8 × 1 MUX using 2 × 1 MUX.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Design and explain the operational characteristics of Dflip flop.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Design Mod-5 synchronous counter using T- flip flop.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Draw a Serial-In Serial-Out Shift register and explain it.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: (Any two)a. BCD codeb. Status Registerc. Ring Counter

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
